

# End-to-End Food Delivery Analytics

## Executive Summary

### PROJECT OVERVIEW

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This project is an end-to-end data analytics case study built using real-world restaurant data from the Kaggle Swiggy Bangalore Restaurant Dataset, encompassing approximately 8,680 restaurants across 9 cities. The objective is to demonstrate how data-driven decision making can improve customer satisfaction, restaurant performance, and operational efficiency in a highly competitive FoodTech platform.

### BUSINESS PROBLEM

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Food delivery platforms face challenges related to pricing optimization, delivery efficiency, restaurant quality assessment, and customer retention. This project addresses these challenges by analyzing pricing, ratings, cuisine diversity, and delivery time to identify performance drivers. Key objectives include identifying factors influencing popularity, segmenting restaurants based on performance metrics, studying the impact of delivery time on order volume, and supporting data-driven restaurant onboarding and promotional strategies.

### APPROACH

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- **Data cleaning and preprocessing** using Python (removing duplicates, handling missing values in ratings and pricing, standardizing cuisine text, and converting delivery time to numerical format).
- **Feature engineering** for business-ready metrics (creating analytical features like `price_range`, `rating_range`, `delivery_range`, `primary_cuisine`, and `popularity_score` for BI visualization).
- **Exploratory Data Analysis (EDA)** to uncover key trends across geographical regions and restaurant types.
- **SQL-based analytical queries** to extract actionable insights.
- **Power BI dashboard** for executive insights and dynamic visual sales analysis.

### KEY INSIGHTS

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- **Delivery speed has a stronger impact on ratings than price.** Fast delivery restaurants average a 3.8 rating, outperforming moderate/slow deliveries, yet only 21.26% of restaurants on the platform offer fast delivery.
- **Multi-cuisine restaurants attract a wider audience**, show higher popularity, and drive higher order volumes compared to single-cuisine establishments.

- **Budget and mid-range restaurants dominate customer engagement.** In contrast, some premium restaurants suffer from low popularity despite maintaining high customer ratings.
- **High ratings do not always translate to high visibility.** For instance, 2,750 high-rated restaurants remain stuck in the "Medium Popularity" bracket, indicating a significant discovery gap.
- **Operational efficiency varies significantly by geography** (e.g., Ahmedabad leads with 42% fast delivery, while Kolkata lags behind at just 3%).

## BUSINESS IMPACT

The project demonstrates how analytics can support restaurant onboarding, marketing strategy, delivery optimization, and customer experience improvements. The framework is scalable to other cities and platforms.

*This project showcases strong analytical thinking, business understanding, and data storytelling skills suitable for Data Analyst internship roles.*